

Graphical solutions to linear equations and inequalities



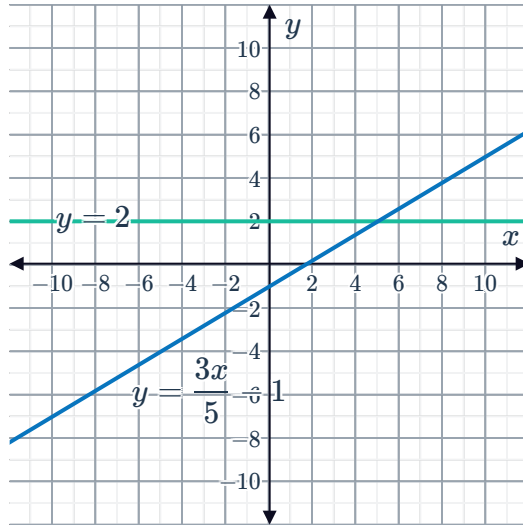
1

a

i $x = 5$

ii $y = 2$ and $y = \frac{3x}{5} - 1$

iii



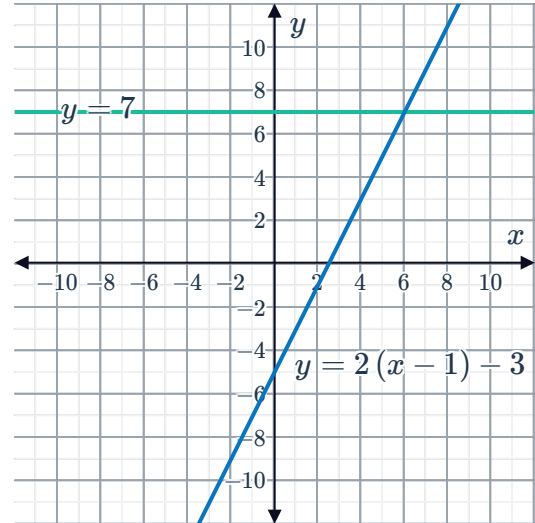
iv $x = 5$

b

i $x = 6$

ii $y = 7$ and $y = 2(x - 1) - 3$

iii



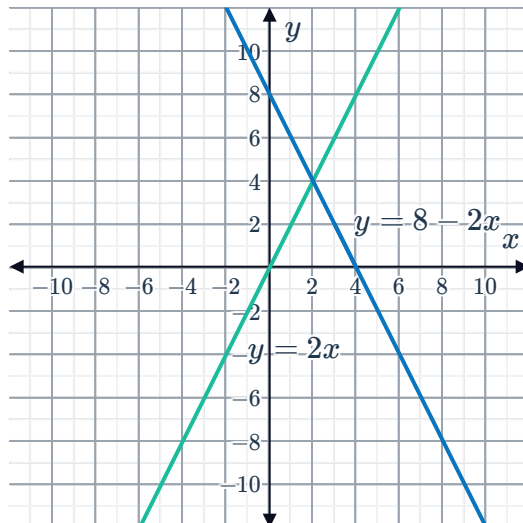
iv $x = 6$

c

i $x = 2$

ii $y = 2x$ and $y = 8 - 2x$

iii



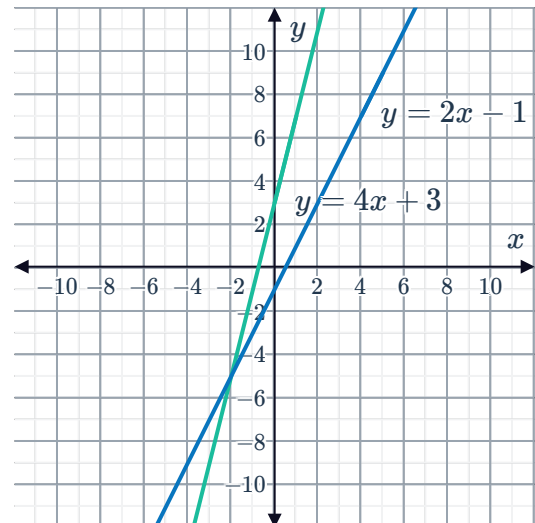
iv $x = 2$

d

i $x = -2$

ii $y = 4x + 3$ and $y = 2x - 1$

iii



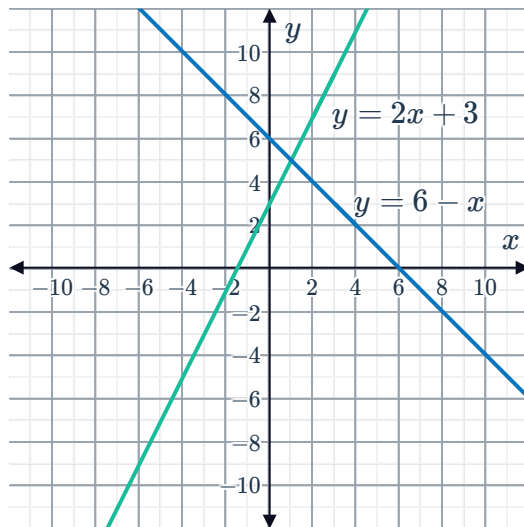
iv $x = -2$

e

i $x = 1$

ii $y = 2x + 3$ and $y = 6 - x$

iii



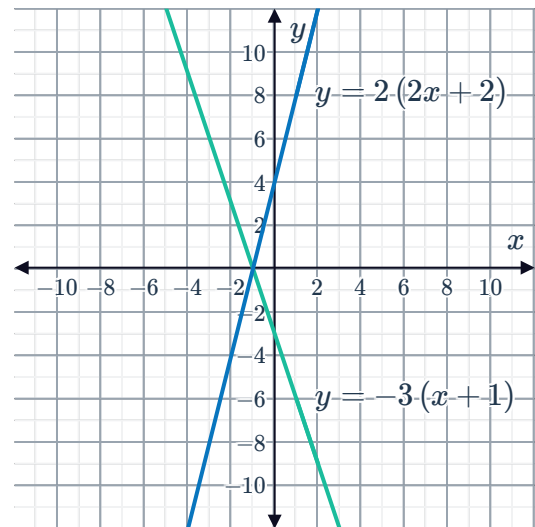
iv $x = 1$



i $x = -1$

ii $y = -3(x + 1)$ and $y = 2(2x + 2)$

iii



iv $x = -1$

2

a $x = 2$

b $x = 2$

c $x = 2$

d $\emptyset$

3

a $x < 6$

b $x \geq -3$

c $x < 5$

d $x > 5$

4 Infinite number of solutions

5 $y = 2x$

6 $y = 4x + 4$

7

a Yes

b No

c No

d No

e Yes

8

a $g(x) = x - 4$

b $x = 0$

9

a



b

 $k = 1$ 